The Quantitative Architect: A Definitive Compendium of Mathematical Aptitude for Advanced Management Entrance

The evolution of high-stakes management entrance examinations, particularly the Common Admission Test (CAT), has transitioned from a traditional assessment of mathematical knowledge to a rigorous evaluation of cognitive agility, structural reasoning, and the ability to apply fundamental principles to multifaceted, non-routine scenarios.[1] In the contemporary testing environment, characterized by a condensed format of 66 questions and a 120-minute duration, the Quantitative Aptitude (QA) section serves as a primary differentiator for elite candidates.[1] Success in this domain necessitates a comprehensive mastery of a vast mathematical spectrum, beginning with the most elementary numerical foundations and extending to sophisticated number theory and coordinate geometry.[1] The following report provides an exhaustive analysis of the curriculum required to achieve total proficiency in CAT-level examinations, articulating the conceptual origins, mechanisms, and strategic implications of every necessary topic.

The Genesis of Numeracy: Foundational Principles and Mental Agility

The bedrock of mathematical aptitude is not the complex formula but the intuitive understanding of number properties and the ability to manipulate them mentally with high velocity.[5] For the advanced aspirant, the initial phase of preparation involves the establishment of "Mental Math Supremacy," where the reliance on external calculators is eliminated in favor of internalized numerical benchmarks.[4] This stage focuses on the "Language of Numbers," ensuring that the candidate can interpret mathematical relationships as fluently as a primary language.[5]

Numerical Benchmarks and Speed Calculation Frameworks

A critical prerequisite for high-level problem solving is the instantaneous recall of specific numerical relationships that frequently appear in algebraic and arithmetic contexts.[6] This recall functions as the engine of the problem-solving process, allowing the candidate to bypass tedious calculation and focus on the logical "twist" of a question.[5] Mastery begins with the memorization of squares and cubes, which are ubiquitous in volume, area, and compound interest problems.[6] Furthermore, the conversion of reciprocal fractions into percentages is essential for navigating the Data Interpretation and Logical Reasoning (DILR) section, where growth rates and ratios are central.[5]

Numerical Category	Required Proficiency Range	Strategic Application
Integer Squares	1 to 30	Essential for Algebra, Geometry, and SI/CI [4]
Integer Cubes	1 to 20	Vital for 3D Mensuration and Geometric Progressions [4]
Reciprocal Fractions	1/2 up to 1/30	Critical for Percentages, Profit/Loss, and TSD [6]
Powers of 2 and 3	2^1 to 2^{10}, 3^1 to 3^6	Foundational for Number Systems and Logarithms [9]
Multiplicative Tables	1 through 20	Baseline requirement for mental agility and speed [4]

Beyond memorization, the adoption of Vedic mathematical techniques provides a structured approach to rapid multiplication, division, and square root extraction.[6] These methods, such as the "base method" for multiplication or the "duplex method" for squaring, allow for the reduction of multi-step operations into single-line mental processes.[6] The mandate for the candidate is clear: if a simple multiplication like 17×13 cannot be performed mentally in under five seconds, the foundational agility required for advanced algebra has not yet been achieved.[5]

The Topology of the Number Line and Signed Integers

The conceptual journey begins with the number line, a geometric representation that defines the spatial relationship between positive and negative integers.[10] Understanding the number line is not merely a matter of plotting points but involves a deep comprehension of directed distance and the foundational rules of integer arithmetic.[10] Negative integers, often a source of confusion for the uninitiated, represent a reversal of direction or a state of deficit, and their manipulation is governed by rigid sign conventions that must be second nature to the aspirant.[10]

A central concept in this domain is the absolute value, denoted by $|x|$, which is defined as the non-negative distance of a number from the origin on the number line.[12] This definition implies that $|x| = |-x|$, a property that serves as the precursor to understanding modulus

functions and inequalities in advanced algebra.[12] The relationship between distance and difference is perhaps the most profound insight at this level: for any two points a and b, the distance between them is $|a-b|$.[11] This principle is the cornerstone of solving complex inequality problems that involve ranges and boundaries.[12]

Fractions, Decimals, and the Rational Continuum

The transition from integers to rational numbers introduces the complexity of the "part-whole" relationship.[13] Aspirants must master the operations of addition, subtraction, multiplication, and division of fractions, emphasizing the role of the Lowest Common Multiple (LCM) in establishing common denominators.[13] The study of rational numbers further involves understanding decimal representations, which are categorized as terminating or non-terminating but repeating.[18]

The ability to switch fluently between fractions, decimals, and percentages is a hallmark of the advanced candidate.[19] This fluency allows for the selection of the most efficient representation for a given problem; for instance, calculating 12.5% of 800 is significantly faster when 12.5% is recognized as the fraction $1/8$.[19] Furthermore, the comparison of rational numbers through cross-multiplication or base-value alignment is a frequent requirement in competitive testing.[13]

Arithmetic: The Pillar of Competitive Aptitude

In the architecture of competitive examinations, arithmetic is the most heavily weighted section, typically accounting for approximately 35% to 40% of the Quantitative Aptitude paper.[8] Its dominance is explained by its practical application in management and its ability to test logical reasoning within a numerical context.[2] Mastery of arithmetic is non-negotiable, as its concepts are not only tested directly but also serve as the engine for solving Data Interpretation sets.[8]

Percentages and the Dynamics of Base Values

Percentages represent the most versatile tool in the arithmetic kit, serving as the foundation for Profit and Loss, Interest, and growth rate analysis.[8] The study of percentages must move beyond the basic formula to an understanding of "Percentage Dynamics," which includes successive changes and base-value shifts.[5]

The concept of successive percentage change is vital for modeling compounding effects. If a value is increased by $a\%$ and then by $b\%$, the net change is given by the formula

$a + b + \frac{ab}{100}$.[9] This relationship is central to solving problems involving population growth, economic inflation, and compound interest.[8] Furthermore, the distinction between a "percentage point" increase and a "percentage change" is a common trap in testing; an increase from 10% to 15% is a 5 percentage point increase but a 50% percentage increase.[8]

The Commercial Logic of Profit, Loss, and Discount

These topics apply percentage principles to the domain of trade and commerce.[7] Advanced problems in this area often involve multi-layered discount structures or "cascaded" profit chains where multiple intermediaries are involved.[2]

Concept	Definition/Formula	Strategic Insight
Cost Price (CP)	Price at which an item is purchased	The fixed base for all profit/loss calculations [7]
Selling Price (SP)	Price at which an item is sold	Determined by applying profit/loss to the CP [7]
Marked Price (MP)	The "sticker" price on an item	The base for calculating discounts (MRP) [8]
Markup Percentage	$\frac{MP-CP}{CP} \times$	Represents the increase over cost before discounts [8]
Profit/Loss Percentage	$\frac{	SP - CP
Discount Percentage	$\frac{MP-SP}{MP} \times$	Always calculated on the Marked Price [8]

A niche but frequent sub-topic is the "Dishonest Dealer" scenario, which tests the candidate's ability to handle deceptive markups and false weights.[8] In these problems, profit is often derived not just from price manipulation but from the reduction of the effective cost through weight adulteration, requiring a careful alignment of the "price-per-unit-weight" logic.[8]

Interest Frameworks: Linear vs. Exponential Growth

The study of Simple Interest (SI) and Compound Interest (CI) is an exploration of the time value

of money.[4] While SI represents linear growth where interest is earned only on the initial principal, CI represents exponential growth where interest itself earns interest.[4]

In competitive examinations, the focus is often on the difference between SI and CI for the same period and rate, or on the impact of different compounding frequencies (annual, half-yearly, quarterly).[7] The effective rate of interest is a critical concept here; for instance, a 10% annual interest rate compounded half-yearly yields an effective annual rate of 10.25%.[7] Furthermore, the application of CI principles to "depreciation" models—where a value decreases by a fixed percentage over time—is a recurring theme.[8]

Ratio, Proportion, and the Science of Variation

Ratios provide a method for comparing quantities of the same dimension, while proportions establish the equality of two ratios.[8] Mastery of this area involves understanding properties like "Componendo and Dividendo," which allows for the rapid manipulation of fractional equations: if $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.[8]

Variation extends this logic to functional relationships between variables.[8] Direct variation ($y = kx$) and inverse variation ($y = \frac{k}{x}$) are the foundations for understanding physics-based problems in math, such as the relationship between speed and time for a fixed distance.[8] Advanced problems may involve "joint variation," where a variable depends on the product or ratio of multiple other factors.[19]

Averages, Mixtures, and the Alligation Rule

This domain focuses on determining the central value of a data set or the characteristics of a mixture created from different components.[8] The concept of "Weighted Average" is paramount here, as it accounts for groups of different sizes having different mean values: $\text{Average} = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}$.[8]

The "Rule of Alligation" is a specialized tool used to find the ratio in which two ingredients of different prices or concentrations must be mixed to achieve a desired mean price or concentration.[8] This rule can be expressed as:

$$\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{\text{Price of Dearer} - \text{Mean Price}}{\text{Mean Price} - \text{Price of Cheaper}}$$

This logic is extensively applied in problems involving milk and water mixtures, alloy compositions, and even weighted averages of populations.[8]

Time, Speed, and Distance: The Dynamics of Motion

Time, Speed, and Distance (TSD) is arguably the most diverse sub-topic in arithmetic, encompassing everything from basic speed calculations to complex relative motion scenarios.[8] The fundamental relationship $D = S \times T$ is simple, but its applications are nuanced.[7]

- **Relative Speed:** This is the cornerstone of TSD. Objects moving in the same direction have a relative speed of $|S_1 - S_2|$, whereas objects moving in opposite directions have a relative speed of $S_1 + S_2$.[8]
- **Boats and Streams:** Involves moving in a medium that itself has a speed. Downstream speed is $B + S$ and upstream speed is $B - S$, where B is the boat's speed in still water and S is the stream's speed.[8]
- **Trains:** Problems focusing on the time taken to cross a point, a platform, or another train, where the length of the objects must be factored into the total distance.[8]
- **Circular Motion:** A high-level topic where runners on a track meet at various points. Key questions involve finding the time until the first meeting and the number of distinct points on the track where the runners will ever meet.[8]
- **Escalators and Moving Walkways:** A modern application of relative speed where a person walks on a surface that is also moving.[8]

Time and Work: The Logic of Efficiency

Time and Work problems explore the inverse relationship between efficiency and the time taken to complete a task.[8] The "Man-Day" concept is a primary tool here, establishing that the total work done is a product of the number of workers, their efficiency, and the time they spend.[8]

A significant sub-topic is "Pipes and Cisterns," which applies work logic to the filling of tanks.[8] In this context, an inlet pipe has positive efficiency, while a leak or an outlet pipe has negative efficiency.[8] Furthermore, problems involving wages require the candidate to understand that payment is generally proportional to the actual work done by an individual, not just the time they spent on the task.[8]

Number Systems: Theory, Structure, and Modular Arithmetic

The Number System section evaluates a candidate's understanding of the fundamental properties of integers and the mechanics of division.[26] This area is often the source of logic-based "Type in the Answer" (TITA) questions that distinguish the top percentiles.[2]

Structure, Divisibility, and Factors

Aspirants must possess a deep understanding of number types, including prime, composite, even, odd, rational, and irrational numbers.[3] Divisibility rules for numbers like 2, 3, 4, 5, 8, 9, 11, and 13 are foundational shortcuts.[2]

The study of factors and multiples involves more than just finding divisors.[26] Candidates must be able to calculate:

- The total number of factors of a number $N = a^p b^q c^r$ as $(p+1)(q+1)(r+1)$.[19]
- The sum of factors and the product of factors.
- The Highest Common Factor (HCF) and Least Common Multiple (LCM), utilizing the relationship that $HCF \times LCM$ equals the product of the two numbers.[7]
- Trailing zeros in factorials, which is determined by counting the powers of 5 within the factorial product.[6]

Advanced Remainder Theorems and Modular Arithmetic

For complex problems where direct division is impossible, modular arithmetic and specialized theorems are utilized.[26]

Theorem	Mathematical Statement	Strategic Context
Fermat's Little Theorem	$a^{p-1} \equiv 1$	Used when the divisor p is prime [26]
Euler's Totient Theorem	$a^{\phi(n)} \equiv 1$	A generalization for any divisor n [6]
Wilson's Theorem	$(p-1)! \equiv -1$	Relates factorials and remainders for prime p [6]
Chinese Remainder Theorem	Solves simultaneous congruences	Used for finding a number with different remainders [26]

Additionally, the concept of "Cyclicity" is used to find the last digit or the last two digits of large exponential expressions.[2] For example, the powers of 7 follow a four-step cycle in their last digit (7, 9, 3, 1), allowing a candidate to find the last digit of 7^{2024} by identifying its position in

the cycle.[2]

Base Systems and Non-Decimal Arithmetic

Most mathematical operations are performed in base-10 (decimal), but competitive exams test the ability to work in alternative bases like binary (base-2), octal (base-8), or hexadecimal (base-16).[2] Candidates must be able to convert numbers between bases and perform basic arithmetic—addition, subtraction, and multiplication—directly within a non-decimal base system.[6]

The Architecture of Algebra: Equations, Functions, and Graphs

Algebra is the framework of variables and functional relationships, making up roughly 30% of the Quantitative Aptitude syllabus.[3] It demands a transition from numerical calculation to symbolic logic.[22]

Equations, Identities, and Roots

- **Linear Equations:** Solving for unknown variables in single or simultaneous equations. Word problems often involve modeling real-world scenarios, such as age problems or cost-mixture problems.[2]
- **Quadratic Equations:** Standardized as $ax^2 + bx + c = 0$. Candidates must master the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, the sum and product of roots, and the concept of the "Discriminant" to determine the nature of roots (real, equal, or imaginary).[2]
- **Higher-Order Equations:** Utilizing theorems like the Remainder Theorem and the Factor Theorem to solve cubic and quartic equations.[6]
- **Algebraic Identities:** Fluency in expansions such as $(a+b)^2$, $(a-b)^3$, and the critical identity $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$.[7]

Inequalities, Modulus, and Optimization

Inequalities test the ability to define ranges of possible values.[16]

- **Modulus Functions:** Represented by $|x|$, these functions define a non-negative magnitude. Solving equations like $|x-1| + |x+3| = 10$ requires checking different "regions" or "intervals" on the number line where the sign of the expressions within the modulus bars changes.[2]
- **Maxima and Minima:** This involves finding the extreme values of functions. For quadratic

equations $y = ax^2 + bx + c$, the vertex occurs at $x = -\frac{b}{2a}$, representing a minimum if $a > 0$ and a maximum if $a < 0$.[33]

- **AM-GM Inequality:** For any set of non-negative real numbers, the Arithmetic Mean (AM) is always greater than or equal to the Geometric Mean (GM).[33] This is a powerful tool for finding the minimum sum of numbers with a fixed product or the maximum product of numbers with a fixed sum.[33]

Functions, Transformations, and Graphical Analysis

A function is a rule that assigns exactly one output to each input.[15] Advanced candidates must understand:

- **Domain and Range:** The set of valid inputs and the resulting set of outputs.[15]
- **Function Types:** Including periodic, even/odd, composite, and inverse functions.[15]
- **Graph Transformations:** Understanding how the graph of $f(x)$ changes when transformed to $f(x \pm a)$, $f(x) \pm a$, or $|f(x)|$.[15] These transformations are essential for solving visual algebra problems where plotting a quick graph can reveal the number of real roots or intersections.[15]

Logarithms, Surds, and Indices

Logarithms are the inverse of exponentiation and are essential for handling equations where the variable is in the exponent.[4] Key properties like $\log(ab) = \log a + \log b$ and the base-change formula $\log_b a = \frac{\log a}{\log b}$ are frequently tested.[7] Similarly, mastery of "Surds and Indices"—the rules of powers and roots—is necessary for simplifying complex algebraic expressions.[7]

Geometry and Mensuration: Spatial Reasoning and Visualization

Geometry and Mensuration account for approximately 15% to 20% of the examination, testing Euclidean principles and 3D spatial analytics.[3]

Euclidean Geometry of the Plane

The study of 2D shapes begins with lines and angles (parallel lines, transversals) and moves into the properties of triangles, circles, and polygons.[3]

- **Triangles:** The most critical shape in geometry. Aspirants must master:
  - **Theorems:** Pythagoras Theorem, Thales Theorem (Basic Proportionality), and the

Angle Bisector Theorem.[21]
  - **Properties:** Congruence (SSS, SAS, ASA, RHS) and Similarity (AAA, SAS, SSS). In similar triangles, the ratio of areas is the square of the ratio of their corresponding sides.[21]
  - **Points of Concurrency:** Incenter, Circumcenter, Centroid, and Orthocenter, along with their associated properties and ratios.[33]
  - **Mass Point Geometry (MPG):** A physics-based technique used to solve complex ratio problems in triangles involving intersecting cevians.[41]
- **Circles:** Understanding tangents, chords, secants, and cyclic quadrilaterals.[4]
- **Polygons:** Interior and exterior angle properties. For a regular n-gon, each interior angle is $\frac{(n-2)\times 180^\circ}{n}$ and the number of diagonals is $\frac{n(n-3)}{2}$.[4]

Mensuration: Area, Surface Area, and Volume

Mensuration is the quantitative study of shapes.[2] While 2D mensuration focuses on area and perimeter, 3D mensuration focuses on surface area (curved and total) and volume.[4]

3D Shape	Volume Formula	Total Surface Area (TSA)
Cube	a^3	$6a^2$
Cuboid	lbh	$2(lb+bh+hl)$
Right Circular Cylinder	$\pi r^2 h$	$2\pi rh + 2\pi r^2$
Right Circular Cone	$\frac{1}{3}\pi r^2 h$	$\pi rl + \pi r^2$ ($l = \sqrt{r^2+h^2}$)
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$

Advanced problems may involve "Frustums" (a cone with the top cut off) or the nesting of shapes, such as finding the volume of the largest cube that can be carved out of a sphere of

radius R.[4]

Coordinate Geometry

This topic translates geometric properties into algebraic equations on a Cartesian plane.[4]

- **Basics:** Distance formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and Section formula for internal/external division.[7]
- **The Straight Line:** Mastery of various forms like slope-intercept ($y = mx + c$), point-slope, and two-point forms.[7]
- **Slopes:** Parallel lines have $m_1 = m_2$, and perpendicular lines have $m_1 m_2 = -1$.[33]
- **Geometry of the Plane:** Area of a triangle given its vertices and the distance of a point from a line.[33]

Modern Mathematics: Combinatorics, Probability, and Progressions

Modern Math represents the frontier of logical counting and sequence analysis.[2]

Permutations, Combinations, and Counting

- **Permutations (^nP_r):** The number of ways to arrange r items from n distinct items, where order matters.[3] Includes circular permutations where the number of arrangements is $(n-1)!$.[3]
- **Combinations (^nC_r):** The number of ways to select r items from n, where order does not matter.[3]
- **Fundamental Principles:** The "Addition Rule" for mutually exclusive options and the "Multiplication Rule" for independent successive steps.[42]
- **Advanced Scenarios:** Counting paths on a grid, the partition of identical items into distinct groups (stars and bars), and derangements (where no item is in its original place).[3]

Probability and Stochastic Models

Probability measures the likelihood of an event occurring within a defined sample space.[7]

- **Conditional Probability:** The likelihood of event A occurring given that event B has already occurred.[42]
- **Independence:** Events that do not influence each other's outcomes.[42]
- **Bayes' Theorem:** A way of updating the probability of a hypothesis based on new

evidence.[42]

- **Geometric Probability:** Using lengths, areas, or volumes to find probabilities in a continuous sample space.[37]

Set Theory and Venn Diagrams

Set theory provides a framework for organizing data into distinct groups.[21] Venn diagrams are the primary visual tool for solving problems involving overlapping categories.[3]

- **Sets of 2 and 3:** Mastery of the principle of inclusion-exclusion to find the size of the union of multiple sets.[7]
- **Logic Sets:** Identifying the number of elements that belong to "exactly one," "exactly two," or "all three" categories in a three-set Venn diagram.[3]

Sequences, Series, and the Binomial Theorem

- **Arithmetic Progression (AP):** A sequence with a constant common difference (d). Sum of n terms is $S_n = \frac{n}{2}(2a + (n-1)d)$.[3]
- **Geometric Progression (GP):** A sequence with a constant common ratio (r). The sum of an infinite GP is $S_\infty = \frac{a}{1-r}$ for $|r| < 1$.[3]
- **Harmonic Progression (HP):** A sequence where the reciprocals of the terms form an AP.[3]
- **Binomial Theorem:** Used for expanding $(x+y)^n$. Mastery involves finding the general term $T_{r+1} = nCr \cdot x^{n-r} \cdot y^r$, the middle term, and the sum of binomial coefficients (2^n).[6]

Integrated and Miscellaneous Topics

Beyond the core branches, several niche areas are tested that require specific logic or specialized formulas.[4]

Trigonometry: Ratios, Identities, and Heights

While not as dominant as algebra or arithmetic, trigonometry often appears in integrated geometry problems.[4]

- **Ratios:** Understanding Sine, Cosine, Tangent, and their reciprocals within the context of a right-angled triangle.[7]
- **Values:** Recall of values for standard angles ($0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$).[48]
- **Identities:** Basic Pythagorean identities such as $\sin^2\theta + \cos^2\theta = 1$ and

$1 + \tan^2 \theta = \sec^2 \theta$.[7]

- **Heights and Distances:** Utilizing angles of elevation and depression to solve for unknown lengths in scenarios involving buildings, poles, or moving objects.[24]

Applied Statistics: Center and Spread

Business aptitude requires a foundational grasp of descriptive statistics.[25]

- **Measures of Central Tendency:** Mean (average), Median (middle value), and Mode (most frequent value).[4]
- **Measures of Dispersion:** Understanding how data is spread. This includes Range, Variance, and Standard Deviation ($\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$).[52]
- **Data Interpretation:** Reading skewed vs. symmetric distributions and understanding the impact of "outliers" on different measures of average.[52]

Temporal and Financial Logic

- **Clocks:** Problems centered on the relative speed of the hour hand (0.5° per minute) and minute hand (6° per minute). Typical questions involve finding the angle between hands at $4:20$ or calculating when the hands will coincide.[24]
- **Calendars:** Using the concept of "Odd Days" to find the day of the week for a specific date or to calculate leap year impacts.[24]
- **Stocks and Shares:** Understanding dividend yields and market-to-face value ratios. For example, a "10% stock at 120" means a share with face value 100 pays a 10 rupee dividend but must be bought for 120 rupees.[55]

Strategic Pedagogy: The Path to Quantitative Supremacy

The magnitude of the Quantitative Aptitude syllabus necessitates a structured, phased preparation strategy.[6] Knowledge of formulas alone is insufficient; CAT tests the ability to apply these formulas under pressure in unfamiliar contexts.[2]

The Three-Phase Evolution

1. **Phase I: Conceptual Foundation (Basics):** Revisiting high-school math (Classes 8-10) to clarify first principles.[2] During this phase, the aspirant builds a "Formula Cheat Sheet" and masters mental calculation shortcuts.[2]
2. **Phase II: Accuracy and Depth:** Practice moves from simple concepts to

medium-difficulty problems.[2] The objective here is to achieve 80% to 90% accuracy on topic-specific tests without a strict time limit.[2]

3. **Phase III: Speed and Strategy:** Introduction of timed sectional mocks.[47] Aspirants learn to identify "Easy" vs. "Hard" questions—a critical skill given the negative marking in MCQs.[2] The goal is to maximize attempts on high-ROI topics like Arithmetic and Algebra while selectively picking easier questions from Geometry and Number Systems.[2]

Ultimately, the mastery of CAT Quants is less about learning "advanced math" and more about learning "smart math"—developing a toolkit of shortcuts, logical traps, and mental agility that allows a candidate to solve 22 to 24 complex problems in 40 minutes.[1] By treating mathematics as a language and consistently practicing the logic of every sub-topic, an aspirant can transform this challenging section into their strongest competitive advantage.[2]

Works cited

1. CAT Syllabus 2025 for MBA: Section Wise Topics, Download PDF - Testbook, accessed on March 29, 2026, https://cdn.testbook.com/1750335101851-CAT%20Syllabus%202025%20for%20MBA_%20Section%20Wise%20Topics%2C%20Download%20PDF.pdf/1750335103.pdf
2. CAT Quantitative Aptitude Syllabus 2026: Topics, Pattern & Smart ..., accessed on March 29, 2026, https://www.careerlauncher.com/blogs/cat-mba/cat-quantitative-aptitude-syllabus/
3. CAT Syllabus 2026: Sections Wise Topics & Weightage - IMS, accessed on March 29, 2026, https://www.imsindia.com/blog/cat/cat-syllabus/
4. CAT Quant Syllabus 2025 with Topic wise weightage - Shiksha.com, accessed on March 29, 2026, https://www.shiksha.com/mba/articles/cat-quantitative-aptitude-syllabus-blogId-196458
5. study.html
6. Important Quant topics for CAT - CATKing, accessed on March 29, 2026, https://catking.in/important-quant-topics-for-cat
7. Important CAT Formulas - BYJU'S, accessed on March 29, 2026, https://byjus.com/free-cat-prep/important-cat-formulas/
8. CAT Quantitative Aptitude Syllabus 2026 PDF Download Subject Wise, accessed on March 29, 2026, https://www.toprankers.com/cat-quantitative-aptitude-syllabus
9. CAT 2025 Important Quant Formulas - Career Launcher, accessed on March 29, 2026, https://www.careerlauncher.com/blogs/cat-mba/cat-quant-formula-pdf/
10. Absolute value - Mathplanet, accessed on March 29, 2026, https://www.mathplanet.com/education/pre-algebra/explore-and-understand-integers/absolute-value
11. Light & Salt Learning - Intro to Integers, Absolute Value, and Distance - Sign in,

accessed on March 29, 2026, https://sites.google.com/view/gedmathcrashcourse/ged-math-crash-course/0-crucial-ged-basics/intro-to-integers-absolute-value-and-distance
12. Teaching Absolute Value of a Number in Math | HMH, accessed on March 29, 2026, https://www.hmhco.com/blog/teaching-absolute-value-of-a-number-in-math
13. Basic Math Concepts for Middle School | Acellus Learning System, accessed on March 29, 2026, https://www.science.edu/acellus/course/basic-math-concepts-ms/
14. Absolute Value - Math - Steps, Examples & Questions - Third Space Learning, accessed on March 29, 2026, https://thirdspacelearning.com/us/math-resources/topic-guides/number-and-quantity/absolute-value/
15. Basic Concepts of Functions and Graphs for CAT Exams, accessed on March 29, 2026, https://bodheeprep.com/functions-graphs-concepts-cat-exam
16. Number line, absolute value, inequalities | Arithmetic & algebra | Quantitative reasoning | Achievable GRE, accessed on March 29, 2026, https://app.achievable.me/study/gre-v2/learn/quantitative-reasoning-number-line-absolute-value-inequalities
17. Number and Operations - National Council of Teachers of Mathematics, accessed on March 29, 2026, https://www.nctm.org/Standards-and-Positions/Principles-and-Standards/Number-and-Operations/
18. Grade 7: Numerical Reasoning – Integers, Percentages, Fractions, Decimal Numbers - Standards Mapping - Georgia Math | Khan Academy, accessed on March 29, 2026, https://www.khanacademy.org/standards/GA.Math/7.NR
19. IMS CAT Formula Cheat Sheet | PDF | Percentage | Numbers - Scribd, accessed on March 29, 2026, https://www.scribd.com/document/893729457/formulasss
20. CAT Quant Important Topics 2025, Weightage & Formula PDF - Planet E, accessed on March 29, 2026, https://planete.in/cat-quant-important-topics-2025-weightage-formula-pdf-by-planet-e/
21. CAT Syllabus 2026 - All Important Topics & Preparation Strategy - iQuanta, accessed on March 29, 2026, https://www.iquanta.in/blog/cat-syllabus-2026/
22. Top 100 CAT Quantitative Aptitude (QA) Questions and Answers - Physics Wallah, accessed on March 29, 2026, https://www.pw.live/mba/exams/top-100-cat-qa-questions-and-answers
23. Top 100 CAT Quant Questions 2026: Quantitative Aptitude - LPT EdTech, accessed on March 29, 2026, https://lptedtech.com/blog/cat/practice-questions/quant/
24. Quantitative Aptitude: Topics, Questions and Answers Guide, accessed on March 29, 2026, https://www.jainuniversity.ac.in/blogs/quantitative-aptitude-topics-questions-and-answers
25. CAT Exam Syllabus 2025, Section Wise to Important Topics with Weightage -

Mindworkzz, accessed on March 29, 2026, https://mindworkzz.in/cat/cat-exam-syllabus/
26. Number System for CAT: Questions | Tricks | Concepts - BodheePrep CAT Prep., accessed on March 29, 2026, https://bodheeprep.com/cat-number-system
27. Number Systems Shortcut for CAT Exam - BYJU'S, accessed on March 29, 2026, https://byjus.com/free-cat-prep/cat-number-systems-shortcut/
28. Content Domains for Subject Matter Understanding and Skill in Mathematics, accessed on March 29, 2026, https://www.ctcexams.nesinc.com/content/docs/CSET_Prep/CS_mathematics_SMR.pdf
29. CAT Remainder Theorem Formula PDF, Download Now - Cracku, accessed on March 29, 2026, https://cracku.in/cat-remainder-theorem-formulas-pdf/
30. Wilson Remainder Theorem - Unacademy, accessed on March 29, 2026, https://unacademy.com/content/cat/study-material/data-interpretation-and-logical-reasoning/details-about-wilson-remainder-theorem/
31. Chinese Remainder Theorem | Brilliant Math & Science Wiki, accessed on March 29, 2026, https://brilliant.org/wiki/chinese-remainder-theorem/
32. Chinese Remainder Theorem for CAT | PDF | Number Theory | Mathematics - Scribd, accessed on March 29, 2026, https://www.scribd.com/document/526560685/Chinese-Remainder-s-Theorem
33. Important Formulae_for CAT - Time4education, accessed on March 29, 2026, https://www.time4education.com/CAT_Important_formulaes/CAT_Important_Formulae.asp
34. 8th Grade Math Curriculum - Miacademy, accessed on March 29, 2026, https://miacademy.co/homeschool-curriculum/8th-grade/8th-math/
35. CAT Exam: Maxima & Minima Solved questions, Algebra Quant, LR & DI - Day 19 - MBA, accessed on March 29, 2026, https://bschool.careers360.com/articles/cat-preparation-strategy-day-19-maxima-minima-questions-qa-di-varc-and-lr
36. QM–AM–GM–HM inequalities - Wikipedia, accessed on March 29, 2026, https://en.wikipedia.org/wiki/QM%E2%80%93AM%E2%80%93GM%E2%80%93HM_inequalities
37. CAT Questions - Algebra: Functions, 2IIM CAT 2023 online classes, CAT coaching in Chennai, accessed on March 29, 2026, https://iim-cat-questions-answers.2iim.com/quant/algebra/functions/
38. Essential Trigonometric Identities and Properties Guide - Superprof, accessed on March 29, 2026, https://www.superprof.com/resources/trigonometric-identities-and-properties-guide/
39. Rules and Theorems to Avoid during CAT Prep, accessed on March 29, 2026, https://online.2iim.com/cat-exam/blogs/cat-gyan/rules-and-theorems-to-avoid-during-cat-prep/
40. CAT Geometry Formulas 2026 PDF: Triangles, Circles, Polygons & More - Toprankers, accessed on March 29, 2026, https://www.toprankers.com/cat-geometry-formulas

41. [MPG] Mass Point Geometry : Tricks for Triangles for CAT Exam, accessed on March 29, 2026, https://bodheeprep.com/cat-mass-point-geometry-tricks-triangles
42. Modern Maths for CAT 2026: PnC, Probability, Sets Strategy, Weightage & Practice Guide, accessed on March 29, 2026, https://bschool.careers360.com/articles/cat-2026-modern-math-simplified-3-concepts-behind-all-questions
43. CAT Formulas PDF, Download CAT Quant Formula Sheet PDF - Cracku, accessed on March 29, 2026, https://cracku.in/cat-formulas-pdf/
44. Binomial Theorem - BYJU'S, accessed on March 29, 2026, https://byjus.com/jee/binomial-theorem/
45. Relation between AM, GM and HM - BYJU'S, accessed on March 29, 2026, https://byjus.com/maths/relation-between-am-gm-hm/
46. Binomial Theorem | Formulas | CAT Exam - Unacademy, accessed on March 29, 2026, https://unacademy.com/content/cat/study-material/mathematics/binomial-theorem/
47. CAT Syllabus 2026 | CAT Exam Syllabus for MBA - VerbalHub, accessed on March 29, 2026, https://www.verbalhub.com/cat-syllabus.php
48. The CAT Trigonometry concepts with questions - CATKing, accessed on March 29, 2026, https://catking.in/the-cat-trigonometry-concepts-with-questions
49. Trig Identities | Formula, List & Properties - Lesson - Study.com, accessed on March 29, 2026, https://study.com/academy/lesson/list-of-the-basic-trig-identities.html
50. CAT Trigonometry Questions | Heights and distances - 2IIM CAT Question Bank, accessed on March 29, 2026, https://iim-cat-questions-answers.2iim.com/quant/geometry/trigonometry/trigonometry_3.shtml
51. Heights & Distances and Trigonometry (CAT/CMAT/GRE/GMAT) - YouTube, accessed on March 29, 2026, https://www.youtube.com/watch?v=UTbMIUnHH94
52. Statistics Questions and Answers For CAT Exam, Clocks Aptitude Questions - Careers360- B Schools, accessed on March 29, 2026, https://bschool.careers360.com/articles/cat-preparation-strategy-day-53-qa-di-varc-and-lr
53. CAT Maths Syllabus 2026 [PDF Download] Important Topics - Toprankers, accessed on March 29, 2026, https://www.toprankers.com/cat-maths-syllabus
54. Clocks and calendars Concept for the CAT exam. - CATKing, accessed on March 29, 2026, https://catking.in/clocks-and-calendars-concept-for-the-cat-exam
55. Quantitative Aptitude –Stocks and Shares – Formulas E-book, accessed on March 29, 2026, https://cdn.splessons.com/2020/01/stocks-and-shares-practice-set-1-splessons.pdf
56. I am extremely shit in quant. PLEASE help : r/CATpreparation - Reddit, accessed on March 29, 2026, https://www.reddit.com/r/CATpreparation/comments/1aymokx/i_am_extremely_sh

it_in_quant_please_help/
57. How to handle CAT QUANT preparation for CAT 2025 : r/CATPrep - Reddit, accessed on March 29, 2026, https://www.reddit.com/r/CATPrep/comments/1jovah7/how_to_handle_cat_quant_preparation_for_cat_2025/